Bayesian Network-Based Scenario Generation for Autonomous Systems Testing

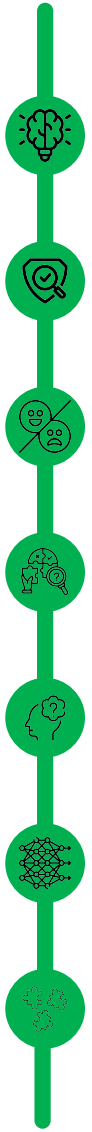
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Overview



Context

Scenario-based testing

Current approaches

Limitations

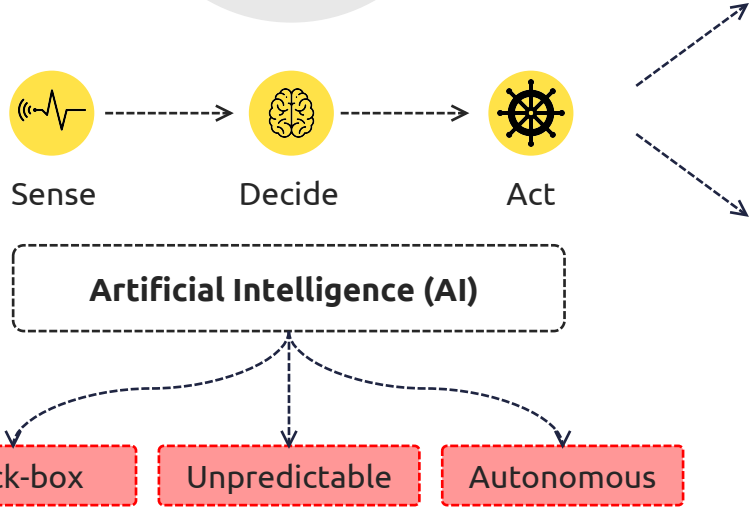
Hypothesis

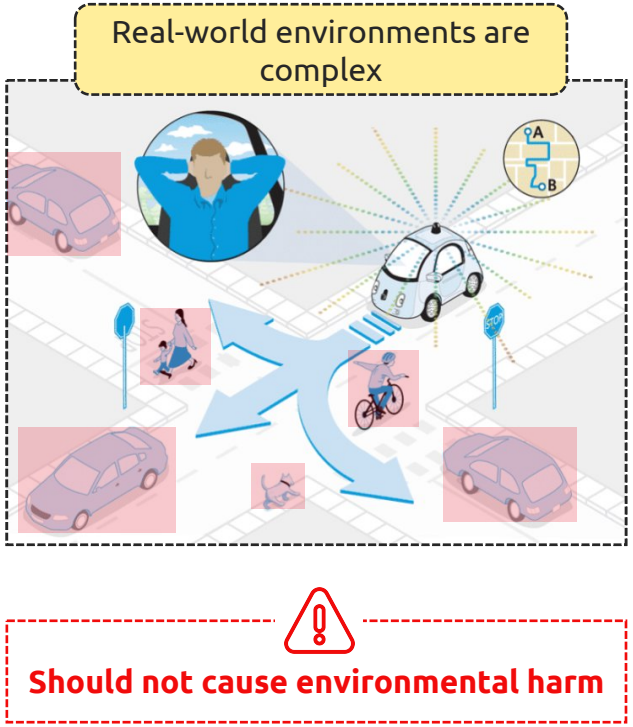
Proposed method

Future Work

Context : Autonomous systems and safety







Safety : the absence of unreasonable (unacceptable) risk. [*ISO 21448*]

Testing is paramount to empirically validating theoretical safety claims

But, how can we test a black-box unpredictable system that interacts with a complex, dynamic, and unpredictable environment?



Scenario-Based Testing as Solution

Real-world conditions

Inspect how the system behave in situations it will encounter in the real world environment

Purposefully designed situations

Focus on safety critical and challenging situations where safety is most vulnerable

System level

Test all system functions working together rather than in isolation, and inspect how the hole system behave

Scenario-based Testing

Scenario-based testing: A method used to test autonomous systems by exposing them to a wide variety of real-world situations, called **scenarios**.

Operational Design Domain (ODD)

The environmental, operational, and behavioral **conditions** under which the system is designed to operate **safely** [*ISO 34503 standard*]

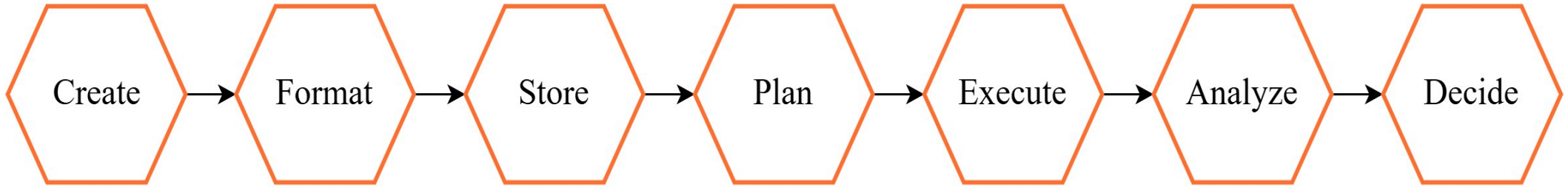
Scenarios

A temporal evolution of scenes, each scene constitutes a static snapshot of the environment, including dynamic agents and static elements

System Under Test (SUT)

Execution environment (**simulator** or in **real-world**)

The Workflow of Scenario-Based Testing



(1) Scenario generation	(2) Scenario execution	(3) Evaluation
Creating scenario content, formatting it into a formalized, machine-readable representation, and organizing these scenarios in a structured database for easy access.	Planning and running the tests in the chosen execution environment.	Analyzing results against pass/fail criteria and determining if safety goals are achieved.

Scenario-based Testing from SOTIF Perspective

Safety of the Intended Functionality (ISO 21448)

SOTIF Primary Focus

SOTIF aims to mitigate risks due to **unexpected operating conditions**. Highlights of this standard include covering:

Insufficient Situational Awareness

Limited perception and understanding of environment

Operational environment challenges

e.g., Weather conditions, infrastructure limitations

Scenario Coverage

emphasis on **enumerating operational scenarios**

SOTIF-aligned scenarios

Minimize unknown hazardous scenarios

High coverage, Exploration of the scenario space

Scalable

Generate scenario suites that represent and model large-scale ODDs or the real-world.

Current Scenario Generation methods

Edge case focused methods

Search-based approaches

Employ fitness functions and optimization algorithms to find critical scenarios.

Boundary-focused method

Identify scenarios at the decision boundaries of the system (between safe and unsafe decisions)

Reinforcement learning

Agents to challenge the system under test

Adept at uncovering critical situations specific to a given System Under Test

Depend on SUT performance; limit exploration.

Sampling methods

Systematically or **randomly** selecting points from the scenario space to create diverse and **representative** scenarios. These methods aim to explore the space efficiently without exhaustively testing all possible combinations.

Ex1: Monte Carlo sampling

Random sampling from specified distributions

Ex2 : Uniform sampling

Values are randomly selected with equal probability across a defined range

Provide the possibility to respect available parameter distributions

Traditional methods assume parameter independence and ignore correlations

Combinatorial testing

Systematically generate scenarios by exploring **combinations** of **key variables** and their possible states

Ex : Pairwise testing

Ensures that every possible pair of variable values appears in at least one scenario

Achieve maximum coverage with minimum scenarios

Fail to capture complexity and correlations; coverage gaps

Data-driven methods

Search-based approaches

leverage historical data and statistical analysis to generate scenarios that reflect patterns, relationships, and dynamics observed in real-world

Captures realistic parameter dependencies and natural probability distributions

Constrained by the limitations of available datasets

Limitations of Current Methods

Generation based on SUT's performance

Edge case focused methods generate scenarios based on the performance of the SUT, but safety validation needs to be seen as a function of the target **ODD**.

Unrealistic parameter distributions

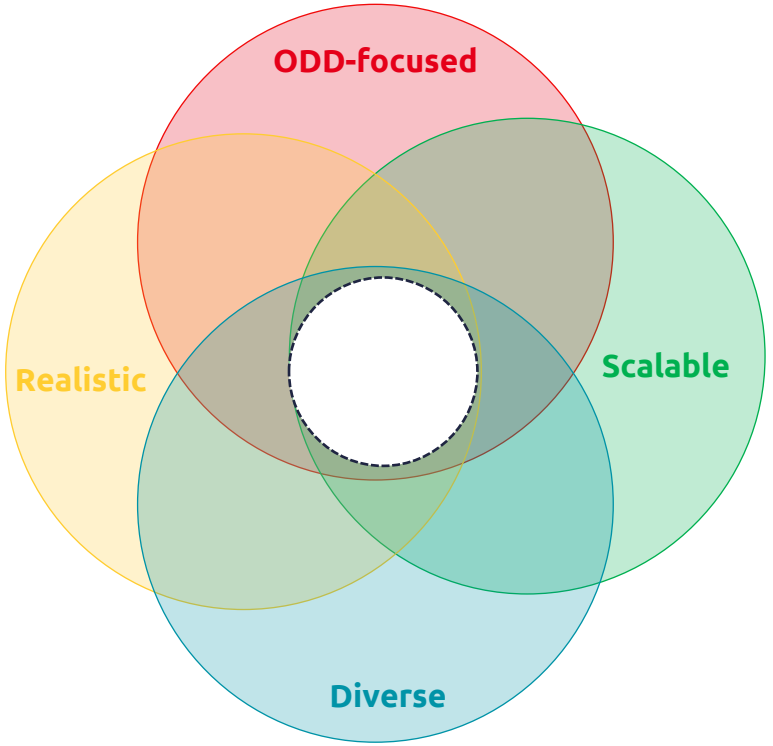
Many approaches do not preserve realistic dependencies and probabilities between scenario parameters.

Incomplete exploration of ODD

Operational Design Domain is rarely explored systematically.

Data limitations

Real-world data is often sparse, biased, or too narrow, constraining scenario diversity.



Research Questions

RQ1: How can we generate **diverse**, **realistic**, and **safety-critical** test scenarios for Autonomous Systems testing?

Coverage

RQ1.1: How do we ensure comprehensive coverage of diverse scenario types?

Realism

RQ1.2: How do we guarantee that generated scenarios reflect real-world driving conditions?

Criticality

RQ1.3: How do we identify and generate scenarios that truly challenge safety boundaries?

RQ2: How can we quantify the risk posed by an autonomous system in different scenarios?

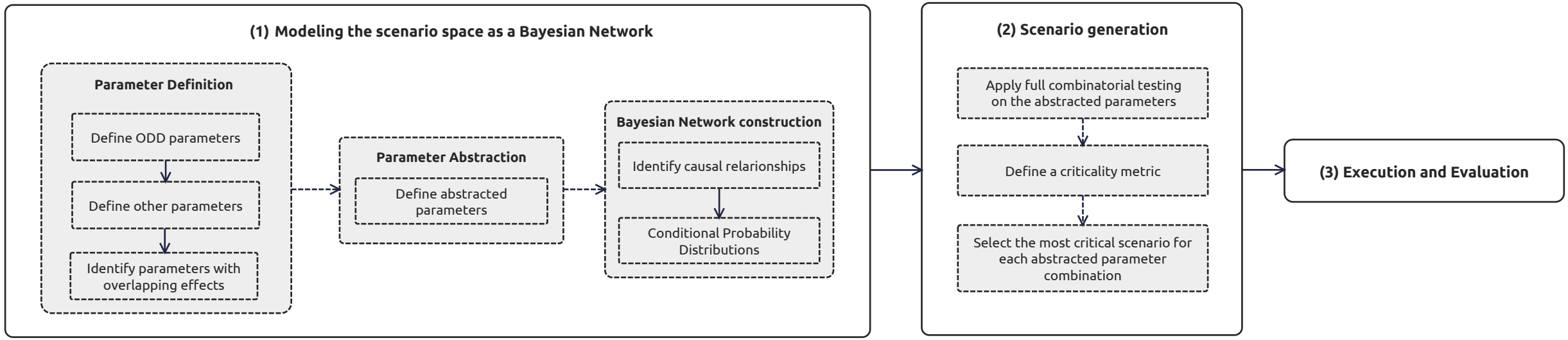
Future research direction



Research Hypothesis

> ***Causal dependencies*** *exist between scenario parameters. By modeling these dependencies with* ***Bayesian Networks***, ***abstracting parameters*** *to reduce dimensionality, and using probabilistic inference to prioritize safety critical scenarios, we can generate more realistic and effective test scenarios for autonomous systems*

Our proposed method - We model before we generate !



Abstracted parameters

Group initial scenario parameteres that have similar effects on the system. For example, **fog density** and **fog distance** influence the system in similar ways (i.e., **visibility**).

Initial Experiments - Evaluating the applicability of our approach

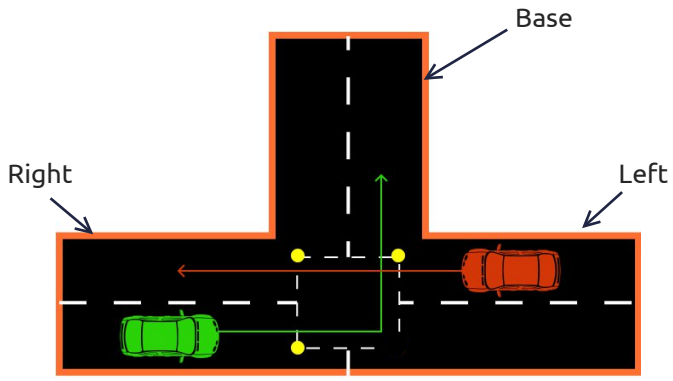
Objectives

1. Evaluate the applicability of our method
2. Compare with existing methods (e.g., Random sampling).

Study Focus

T-Junction Scenarios

- One of the highest accident rates
- Challenging and safety-critical situations
- **3** Possible collision points



An example T-junction scenario used in this study, the ego vehicle (green) interacts with an adversarial vehicle (red). Yellow markers indicate potential collision points.

Scenario space

Intersection Parameters

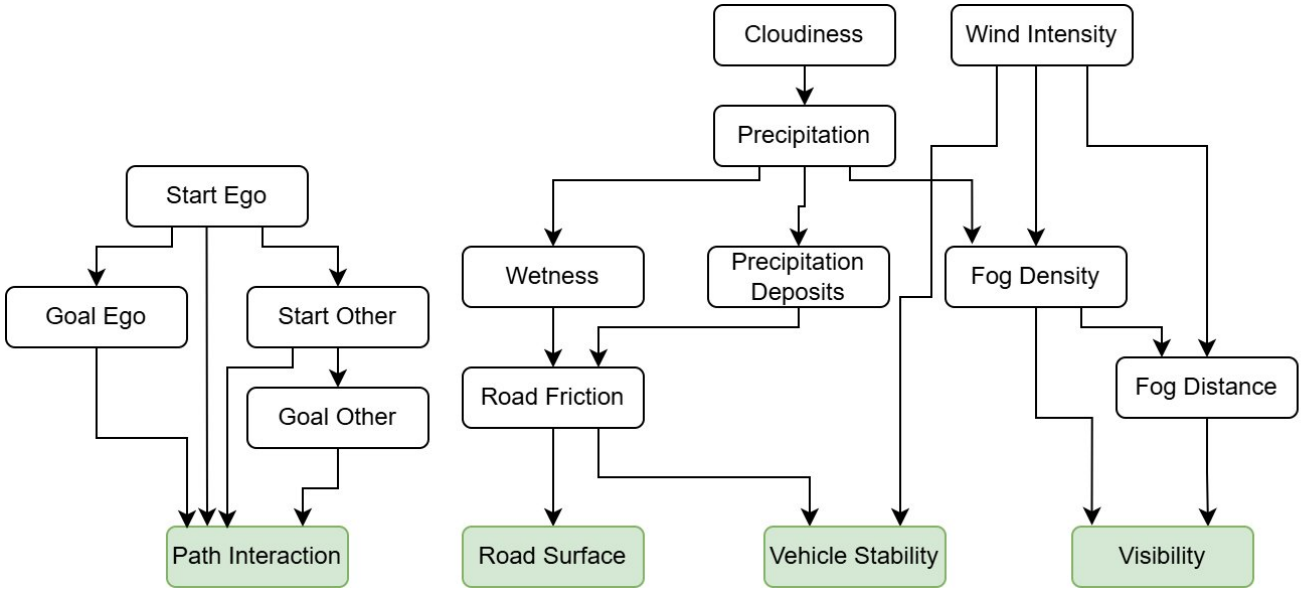
Start Ego	{left, right, base}
Goal Ego	
Start Other	
Goal Other	

Environment Parameters

Road friction	{0, 20, 40, 60, 80, 100}
Fog density	
Precipitation	
Precipitation deposits	
Cloudiness	
Wind intensity	
Wetness	
Fog distance	

$$6^8 * 3^4 = 136\,048\,896$$

LLM Generated Bayesian Network structure & Probability Distribution Functions (PDFs)



The Bayesian network structure, generated with an LLM, with initial parameters depicted in white squares and abstract parameters highlighted in green squares

Cloudiness	Precipitation = 0	20	40	60	80	100
0	0.95	0.05	0.0	0.0	0.0	0.0
20	0.70	0.25	0.05	0.0	0.0	0.0
40	0.40	0.40	0.15	0.05	0.0	0.0
60	0.20	0.30	0.30	0.15	0.05	0.0
80	0.05	0.15	0.30	0.30	0.15	0.05
100	0.0	0.05	0.20	0.35	0.25	0.15

Probability Distribution Function (PDF) for **Precipitation** parameter

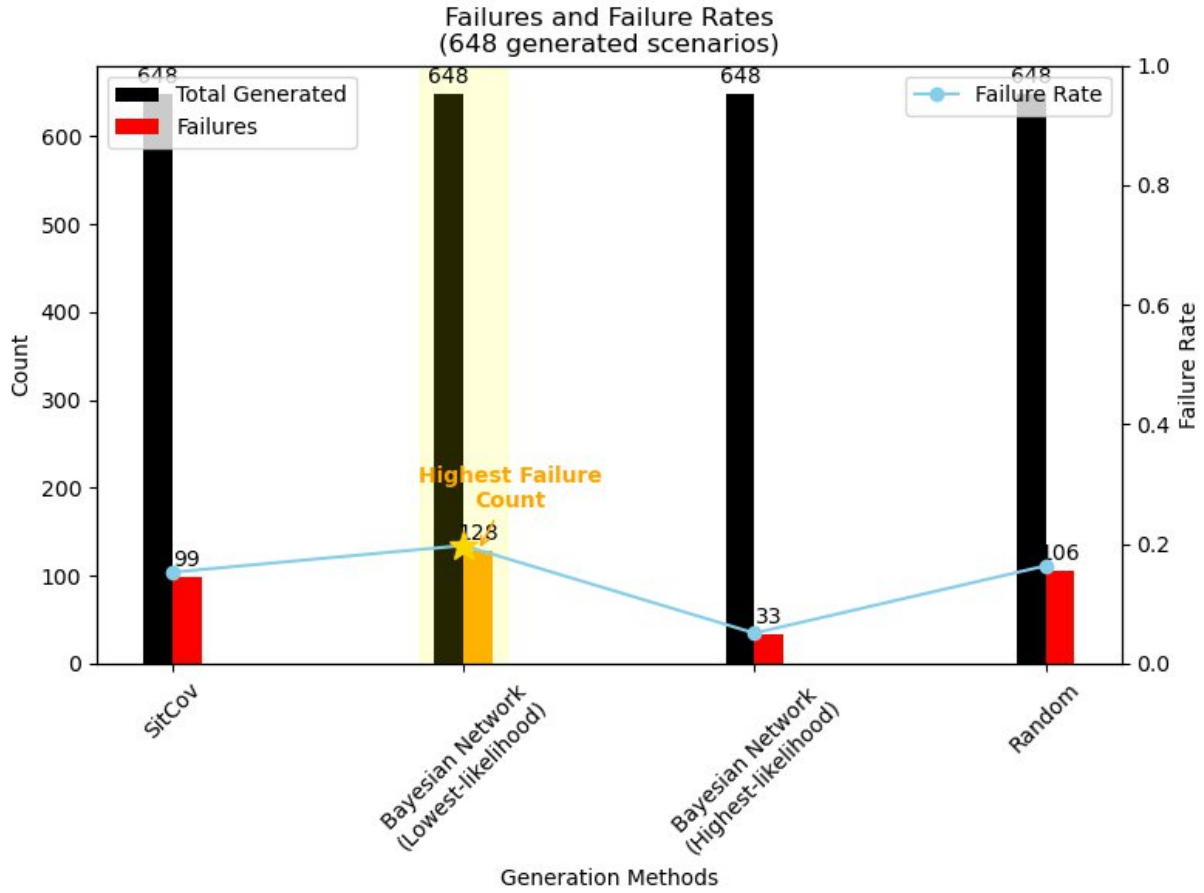
Pruning implausible combinations reduced the scenario space from 136.05 million to **1.75 million** valid combinations → **98.7% reduction**.

Abstracted Parameters	
Path Interaction (Collision point)	**{c1, c2, c3}**
Road Surface	**{0, 20, 40, 60, 80, 100}**
Vehicle Stability	
Visibility	

The total number of scenarios generated from the full combinatorial space of the abstracted parameters is **$6^3 \times 3 = 648$ scenarios**.

Context | Scenario-based testing | Current approaches | Limitations | Hypothesis | Proposed method | Future Work

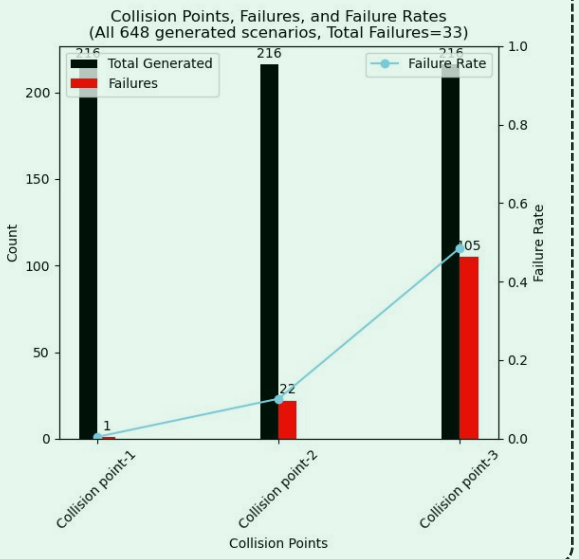
Our approach uncovers more failures than SitCov and Random Sampling



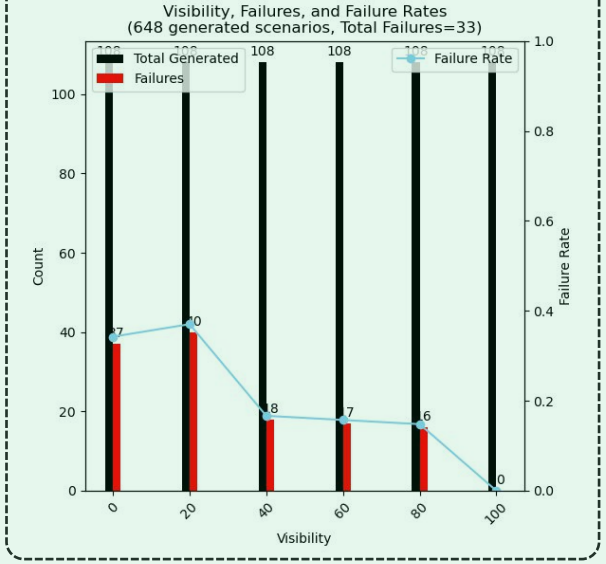
Metric	SitCov	**Bayesian Net (least)**	Bayesian Net (most)	Random
Total failures	99/648 (15.28%)	**128/648 (19.75%)**	33/648 (5.09%)	106/648 (16.36%)

Identifying Triggering Conditions and Revealing System Weaknesses

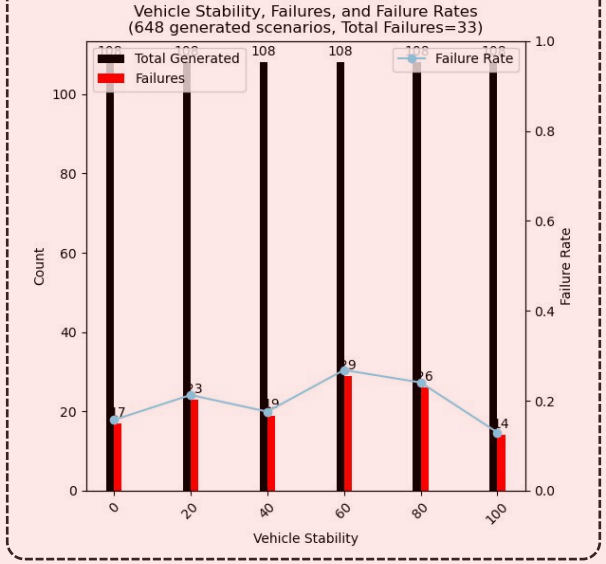
Collision point



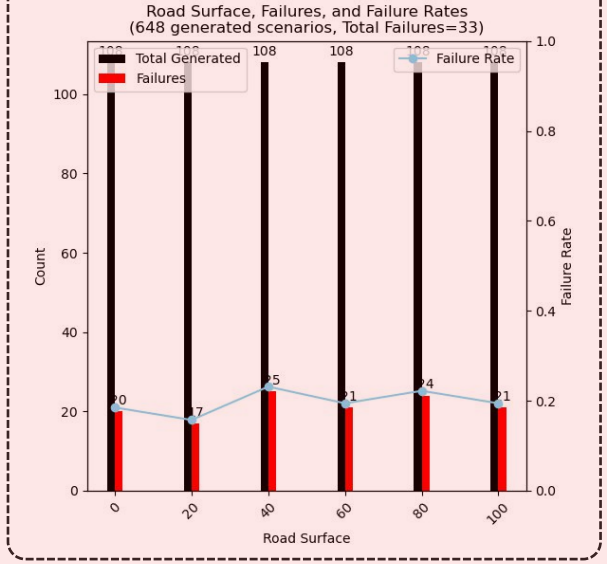
Visibility



Vehicle Stability



Road Surface



Road surface and **vehicle stability** have a uniform failure distribution → no strong influence detected.

Low visibility conditions lead to a significantly higher failure rate.

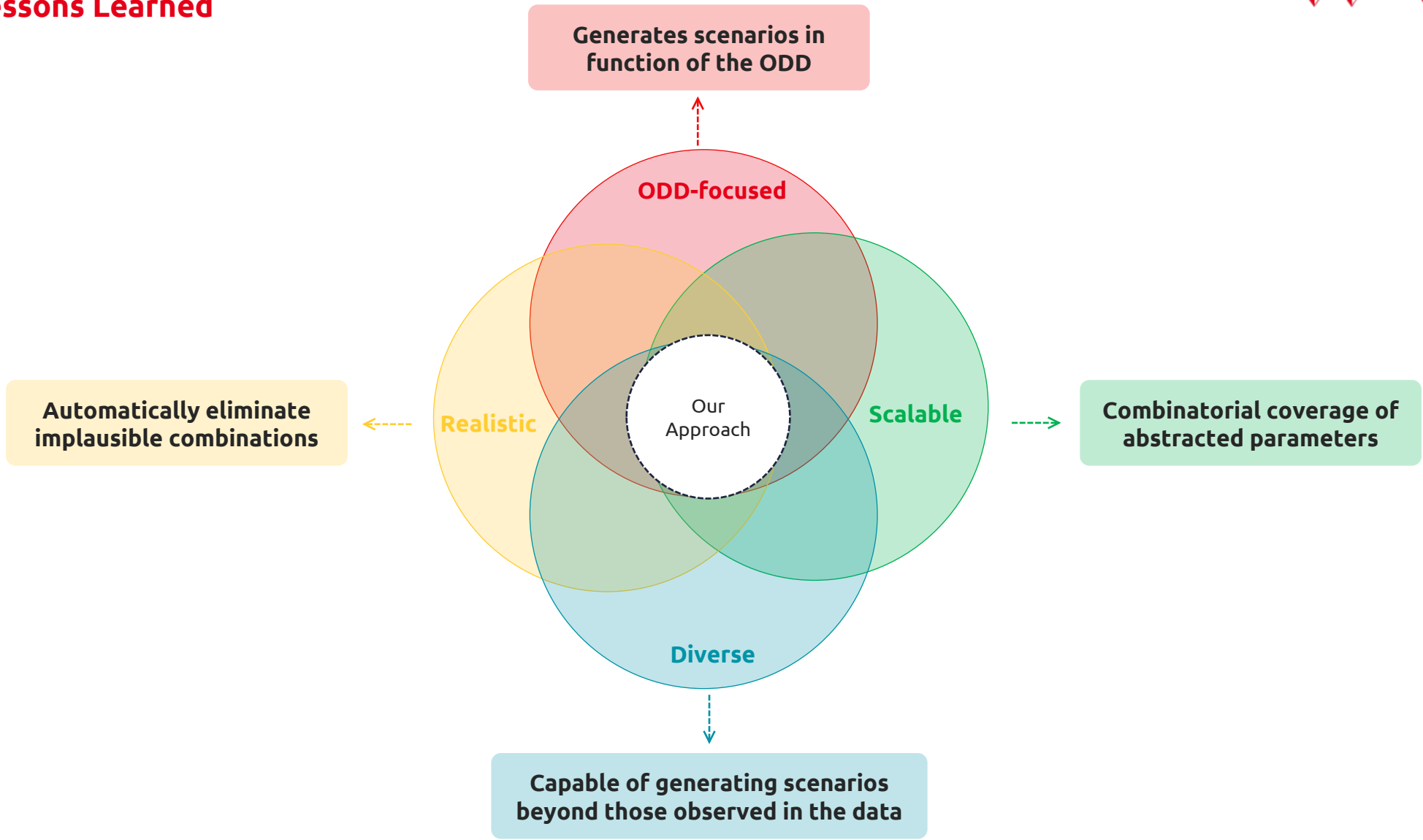
The vehicle struggles more when encountering another vehicle at **collision point c3**.

Collision point and **visibility** are key parameters affecting system behavior → these are **most likely the triggering conditions** for unsafe behavior.

For context, the vehicle model used in the experements relies solely on **one camera-based perception module** → our approach successfully identified its inherent weakness by revealing its sensitivity to low visibility.

Context | Scenario-based testing | Current approaches | Limitations | Hypothesis | Proposed method | Future Work

Summary & Lessons Learned



Positioning our approach

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Achieve maximum coverage with minimum scenarios



Future Work

1. **Generalizability & broader validation**: test the methodology across diverse scenarios and multiple Systems Under Test (SUT) to evaluate its effectiveness and generalizability beyond the current case study.

2. **Incorporating a criticality metric**: Integrate a criticality metric into the framework to capture of safety-critical situations.

3. **Defining safety evaluation metrics** : quantitatively assess whether the SUT meets safety goals.



References

[***Slide3.1***] International Organization for Standardization. Software and systems engineering — Software testing — Part 11: Guidelines on the testing of AI-based systems. Technical Report ISO/IEC TR 29119-11:2020, International Organization for Standardization, 2020.

[***Slide3.2***] ISO 21448:2022. Road vehicles — Safety of the intended functionality. Geneva, Switzerland: International Organization for Standardization (ISO), 2022.

[***Slide6***] Xizhe Zhang, Siddartha Khastgir, Justin-Kiyoshi Tiele, Kazuhito Takenaka, Tasuku Hayakawa, and Paul Jennings. Odd and behavior based scenario generation for automated driving systems. IEEE Access, 12:10652–10663, 2024.

[***Slide7.1***] Lukas Birkemeyer, Christian King, and Ina Schaefer. Is scenario generation ready for sotif? a systematic literature review. In 2023 IEEE 26th International Conference on Intelligent Transportation Systems (ITSC), pages 472–479. IEEE, 2023.

[***Slide7.2***] Lukas Birkemeyer, Tobias Pett, Andreas Vogelsang, Christoph Seidl, and Ina Schaefer. Feature-interaction sampling for scenario-based testing of advanced driver assistance systems. In Proceedings of the 16th International Working Conference on Variability Modelling of Software-Intensive Systems, pages 1–10, 2022.

[***Slide8.1***] Hans J He, Daniel J Stilwell, Mazen Farhood, and Devaprakash Muniraj. Use of falsification to find rare failure modes of a ship collision avoidance algorithm. In OCEANS 2022-Chennai, pages 1–9. IEEE, 2022.

[***Slide8.2***] Yiming Xu, Yajie Zou, and Jian Sun. Accelerated testing for automated vehicles safety evaluation in cut-in scenarios based on importance sampling, genetic algorithm and simulation applications. Journal of intelligent and connected vehicles, 1(1):28–38, 2018.

[***Slide8.3***] Xudong Zhang and Yan Cai. Building critical testing scenarios for autonomous driving from real accidents. In Proceedings of the 32nd ACM SIGSOFT International Symposium on Software Testing and Analysis, pages 462–474, 2023.

[***Slide8.4***] Hong Shu, Haoran Lv, Kang Liu, Kang Yuan, and Xiaolin Tang. Test scenarios construction based on combinatorial testing strategy for automated vehicles. IEEE Access, 9:115019–115029, 2021.

[***Slide9***] Patrick Weissensteiner, Georg Stettinger, Siddartha Khastgir, and Daniel Watzenig. 2023. Operational design domain-driven coverage for the safety argumentation of automated vehicles. IEEE Access 11 (2023), 12263–12284.

[***Slide11***] UC Berkeley. (n.d.). **CS188**: *Introduction to Artificial Intelligence*. Retrieved from http://ai.berkeley.edu

[***Slide13, Slide17***] Zaid Tahir and Rob Alexander. Intersection focused situation coverage-based verification and validation framework for autonomous vehicles implemented in CARLA. In Modelling and Simulation for Autonomous Systems, pages 191–212. Springer, 2022.

Thank you

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